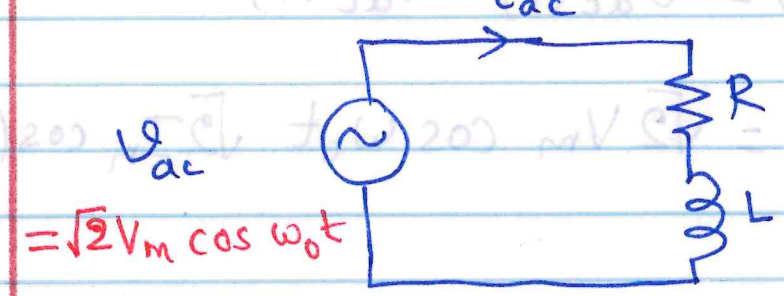
Lecture 3. ECE 330



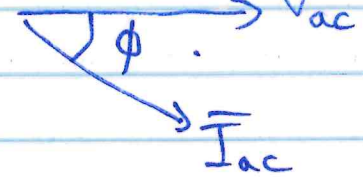
$$\bar{v}_{ac} = V_m \angle 0^\circ = V_m e^{j0}$$

$$\bar{i}_{ac} = I_m \angle \phi = I_m e^{j\phi}.$$

$$\bar{v}_{ac} = \bar{i}_{ac} \cdot (R + j\omega_0 L).$$

$$V_m \angle 0^\circ = I_m \angle \phi \, [R + j\omega_0 L]$$

$$\boxed{\phi = \tan^{-1} \frac{\omega_0 L}{R}} \qquad I_m \angle \phi = \frac{V_m \angle 0^\circ}{R + j\omega_0 L}$$



$$= \frac{V_m}{\sqrt{R^2 + \omega_0^2 L^2}} \cdot \angle -\tan^{-1}\left(\frac{\omega_0 L}{R}\right)$$

ϕ = lagging

$$i_{ac} = \frac{\sqrt{2} V_m}{\sqrt{R^2 + \omega_0^2 L^2}} \cos(\omega_0 t - \phi)$$

I_m

$$p(t) = v_{ac}(t) \cdot i_{ac}(t)$$

$$= \sqrt{2} V_m \cos \omega_0 t \; \sqrt{2} I_m \cos(\omega_0 t - \phi)$$

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$p(t) = \sqrt{2} V_m \cos \omega_0 t \cdot \sqrt{2} I_m \cos(\omega_0 t - \phi)$$

$$= \frac{2}{2} V_m I_m \left[\cos(2\omega_0 t - \phi) + \cos \phi \right]$$

$$= V_m I_m \left[\cos \phi + \cos(2\omega_0 t - \phi) \right]$$

$$= V_m I_m \left[\cos\phi + \cos 2\omega_0 t \cos\phi + \sin 2\omega_0 t \sin\phi \right]$$

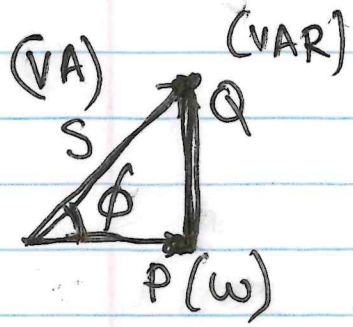
$$p(t) = V_m I_m \cos\phi \left[1 + \cos 2\omega_0 t \right] + \boxed{V_m I_m \sin\phi} \sin 2\omega_0 t$$

$$\langle p(t) \rangle_T = \frac{1}{T} \int_0^T p(t)\, dt = V_m I_m \cos\phi$$

$$P = V_m I_m \cos\phi$$

$$Q = V_m I_m \sin\phi$$

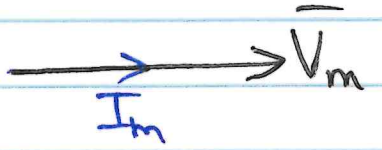
$$P^2 + Q^2 = \underbrace{(V_m I_m)^2}_{S}$$



R

$\phi = 0$

Phasor



Power Triangle



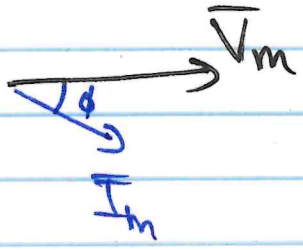
$Q = 0$

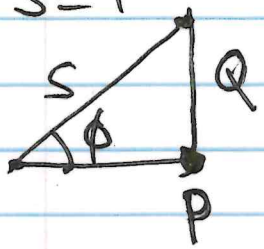
$S = P$

RL

ϕ = negative

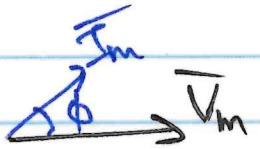
current lags voltage

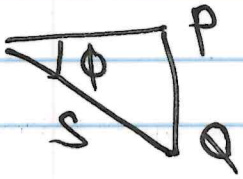




RC

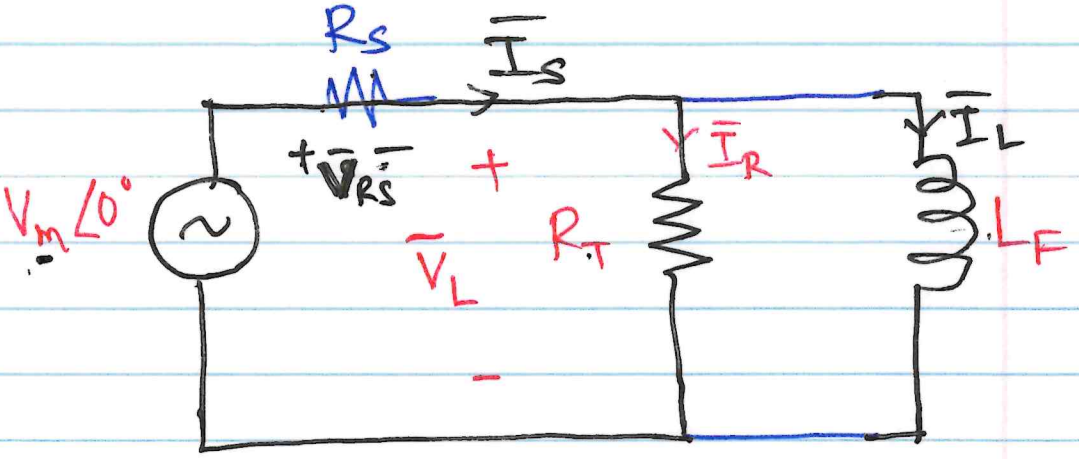
ϕ = current leads voltage





$\cos\phi$ = Power factor

Luke.



$$\bar{I}_R = \frac{\bar{V}_L}{R} \qquad \bar{I}_L = \frac{\bar{V}_L}{j\omega_0 L_F}$$

$$= \frac{\bar{V}_L}{\omega_0 L_F} \angle -90°$$

$$\bar{I}_S = \bar{I}_L + \bar{I}_R \quad \text{KCL}$$

$$V_m \angle 0° = \bar{V}_{RS} + \bar{V}_L \quad \text{KVL}$$

$$P_{loss,\,R_S} = \bar{V}_{R_S} \bar{I}_S$$

$$= (R_S \bar{I}_S)\,\bar{I}_S$$

$$= R_S (\bar{I}_S)^2 = |\bar{I}_S|^2 R_S$$